

TECHNOLOGICAL DESIGN

Technological design courses provide students with a variety of learning experiences that focus on the practical application of the principles of engineering, architecture, and design. These activity-based courses emphasize problem solving to meet design challenges in a wide range of areas, which may include apparel and textile design, architectural design, interior design, mechanical and industrial design, and robotics and control systems.

Students learn to apply knowledge of research, historical trends, design, materials, fabrication methods, and testing criteria to develop innovative and environmentally sustainable products, processes, and/or services. The technologies and processes used to create design solutions may include both traditional and computer-based drafting methods, scale models, working prototypes, animations and simulations, displays, portfolios, and presentations.

The list of approved emphasis areas for technological design can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Technological Design, Grade 12

University/College Preparation

TDJ4M

This course introduces students to the fundamentals of design advocacy and marketing, while building on their design skills and their knowledge of professional design practices. Students will apply a systematic design process to research, design, build, and assess solutions that meet specific human needs, using illustrations, presentation drawings, and other communication methods to present their designs. Students will enhance their problem-solving and communication skills, and will explore career opportunities and the postsecondary education and training requirements for them.

Prerequisite: Technological Design, Grade 11, University/College Preparation

A. TECHNOLOGICAL DESIGN FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of criteria, relationships, and other factors that affect technological design and the design process;
- A2.** describe strategies, techniques, and tools for researching, organizing, planning, and managing design projects and related activities, with an emphasis on advocacy, diplomacy, and marketing;
- A3.** demonstrate an understanding of drafting standards, drawing types, conventions, and guidelines used when representing design ideas graphically;
- A4.** demonstrate an understanding of various types of models and prototypes, and describe the tools, materials, equipment, and processes for building, testing, and evaluating them;
- A5.** use appropriate technical language and communications methods to document, report, present, and market design ideas and results.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe environmental and societal needs (e.g., *barrier-free access, alternative-energy vehicles*) that influence product designs, with reference to key technological concepts (e.g., *aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) (see pp. 7–8);
- A1.2** describe how the results of each step in the design process affects the next step in the process (e.g., *prototype testing can show that more idea development is needed*) (see pp. 22–23);
- A1.3** identify and establish design criteria (e.g., *style, aesthetics, functionality, cost, market*) for a variety of clients and environments (e.g., *business, health care, entertainment, religious*), based on interviews with clients, technical requirements, and research.

A2. Research and Project Management

By the end of this course, students will:

- A2.1** identify and locate sources of technical data and related information for a design project (e.g., *trade literature, catalogues, applicable codes,*

municipal and provincial laws and regulations, workshops, seminars), using a variety of techniques and tools;

- A2.2** describe strategies for organizing, planning, and managing the human, material, and financial resources for a design project and related activities, with an emphasis on advocacy of design ideas and rationales, diplomacy in dealing with clients and suppliers, and marketing of design solutions (e.g., *establishing roles for project team members, advocating for environmentally sound materials, promoting and marketing innovative designs*).

A3. Representing Design Ideas Graphically

By the end of this course, students will:

- A3.1** compare different methods for representing design ideas graphically (e.g., *annotated sketches, expressive drawings, design layouts, computer-aided drafting*), and identify examples of best practices;
- A3.2** demonstrate an understanding of drawing types (e.g., *2D and 3D drawings, floor plans, elevations, sections, detail drawings, rendered drawings*), and of drafting standards, conventions, and guidelines (e.g., *dimensions, symbols, abbreviations, geometries, tolerances; standards and*

guidelines provided by the American National Standards Institute [ANSI] and the American Society of Mechanical Engineers [ASME]);

- A3.3** accurately interpret technical references, drawings, test data, and specifications.

A4. Making and Testing Models and Prototypes

By the end of this course, students will:

- A4.1** compare a variety of types of models and prototypes (*e.g., models: conceptual, physical, virtual, theoretical; prototypes: proof-of-principle, functional, form study*) and modelling tools, equipment, materials, and procedures (*e.g., tools: foam-injection moulding machine; rapid prototyper; computer-aided mill, lathe, or sewing machine; materials: metals, thermoplastics, neoprene*) in terms of suitability, time, budget, and availability;
- A4.2** identify and compare tools and equipment used to assess models and/or modelling processes (*e.g., three-dimensional coordinate measuring machine, decibel meter, height gauge, compression tester*);
- A4.3** describe criteria for assessing models (*e.g., functionality, size, weight, durability, mechanical and physical features*) and modelling processes (*e.g., material costs, assembly time, waste produced*) for a given project.

A5. Reporting and Presenting

By the end of this course, students will:

- A5.1** use technical terminology correctly when documenting design projects, reporting and presenting results, and marketing designs (*e.g., vernier caliper, scale versus ruler, shears versus scissors, geometries versus shapes*);
- A5.2** describe and compare formats and tools for documenting, reporting, presenting, and marketing design ideas and results throughout the design process (*e.g., formats: oral or written presentation, multimedia production, theatrical presentation, role play; tools: multimedia software, display board, image board [lifestyle, mood, styling, usage]*);
- A5.3** describe and compare various reporting styles and formats (*e.g., styles: American Psychological Association [APA], Modern Language Association [MLA]; formats: portfolio, technical report, critique*).

B. TECHNOLOGICAL DESIGN SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use appropriate resources, methods, and tools to research and manage design projects and related activities;
- B2.** apply appropriate methods for generating and graphically representing complex design ideas and solutions;
- B3.** create, test, and analyse models and/or prototypes, using a variety of techniques, tools, and materials;
- B4.** use a variety of formats and tools to create and present reports summarizing and evaluating the design process, to analyse decisions made during the process, and to advocate the final design.

SPECIFIC EXPECTATIONS

B1. Researching and Managing Projects

By the end of this course, students will:

- B1.1** research and analyse pertinent information from appropriate resources (*e.g., building codes, Machinery's Handbook, interviews, union contracts*);
- B1.2** use project management methods and tools (*e.g., checklists, templates, software, surveys, focus groups, questionnaires*) to support, plan, and manage components (*e.g., financial, labour, material*) of design projects and related activities, with a view to incorporating effective advocacy and marketing principles in the process.

B2. Developing and Representing Design Ideas

By the end of this course, students will:

- B2.1** use freehand sketches to help brainstorm initial design concepts for a project;
- B2.2** apply mathematical and scientific concepts and skills as required in the course of designing various products and/or processes;
- B2.3** create presentation-quality drawings to represent design ideas, using a variety of principles and elements of graphic design;

- B2.4** produce a variety of types of hand-drafted and/or computer-based technical drawings (*e.g., elevation, schematic, exploded view*) of design solutions, using industry-recognized drafting standards and conventions (*e.g., geometric tolerancing, schedules*), with an emphasis on illustrations and presentation-quality drawings.

B3. Making and Testing Models and Prototypes

By the end of this course, students will:

- B3.1** select and use appropriate tools, equipment, and materials when creating design models and/or functional prototypes;
- B3.2** use appropriate metric and imperial measuring tools, scales (*e.g., metric: 1:10, 1 cm:1 m or 1:100, 1:500; imperial: 1/2":1' or 1:24*), and proportion techniques when creating and assessing models and/or prototypes;
- B3.3** analyse products and/or processes on the basis of student-justified criteria (*e.g., aesthetics, ergonomics, performance, functionality, cost, environmental stewardship*), with an emphasis on marketability;
- B3.4** finish a model or prototype to professional standards (*e.g., surface finish, detailing, painting*).

B4. Reporting and Presenting

By the end of this course, students will:

B4.1 create and present reports summarizing and evaluating all aspects of the design process, using a variety of tools (*e.g., multimedia equipment, simulation overlays, magnetic illustration board*), with an emphasis on promotional and marketing strategies;

B4.2 report on and analyse decisions made throughout the design process, and advocate for the final design, using a variety of oral and/or written formats.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmentally responsible design practices, and apply them in the technological design process and related activities;
- C2.** analyse the relationship between society and technological development.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** identify and analyse environmental effects of a particular industry or technological system (*e.g., mass transit system, strip mining, sewer system*), and recommend practices that are economically and environmentally sustainable;
- C1.2** describe ways in which environmental issues influence the design of products and/or processes;
- C1.3** describe, advocate for, and apply best practices for conserving energy and other resources when designing a product or process (*e.g., reuse or recycle lumber and other materials; use materials with recycled content; use wood glue instead of hot glue; use renewable energy sources, high-efficiency motors and appliances, and passive heating and cooling of buildings*);
- C1.4** describe ways to reduce the waste produced by the manufacture and use of products (*e.g., cutting patterns that minimize leftover materials, use of materials that are easily recycled, energy-management controls in electronic equipment*), and apply such practices when developing and building prototypes.

C2. Technology and Society

By the end of this course, students will:

- C2.1** independently research and report on political, economic, cultural, and/or environmental issues that affected technological innovations in the past (*e.g., traffic congestion spurred development of compact vehicles, increasing population density led to the construction of taller buildings*);
- C2.2** describe examples of how culture, economics, and politics could influence the future design of products and/or processes (*e.g., environmental awareness and rising costs for fossil fuels could increase the development and use of alternative energy sources*);
- C2.3** describe how technological change affects society (*e.g., developments in telecommunications, health care, and robotics*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** describe and apply personal and environmental health and safety standards and practices related to technological design;
- D2.** compare a variety of careers related to technological design, as well as the training and educational requirements for them, and maintain a portfolio of their work as evidence of their qualifications for future education and employment.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and describe the bodies and agencies that regulate, promote, and test the safety of technological products and/or processes (e.g., *Health Canada, Canadian Society of Safety Engineering [CSSE], Canadian Standards Association [CSA]*), and explain how they work to prevent accidents and enforce standards (e.g., *certification, product recalls*);
- D1.2** adhere to and promote personal and environmental health and safety standards and procedures with respect to processes, materials, tools, equipment, and facilities throughout the design process and when performing related activities (e.g., *use protective equipment; set tool and equipment guards properly; ensure adequate ventilation and ergonomic workplace arrangements; follow safe operating procedures and maintain tools and equipment in good working condition; keep work areas clean and organized; store materials and dispose of wastes properly; report safety violations*);
- D1.3** use protective clothing, gear, and equipment appropriately (e.g., *dust mask, safety glasses*);
- D1.4** describe the rights and responsibilities of employees under the Occupational Health and Safety Act (e.g., *right to know, right to refuse, right to participate*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify and compare a variety of career opportunities related to technological design (e.g., *architect versus architectural technologist, draftsman versus designer*);
- D2.2** compare postsecondary education (e.g., *university, college, skills training centres*) and entry requirements (e.g., *portfolio, internship*) for a variety of technological design careers, and describe lifelong professional development opportunities associated with them (e.g., *continuing education, workshops, seminars*);
- D2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the technological design industry, as identified in the Ontario Skills Passport (e.g., *reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D2.4** demonstrate an understanding of and apply the work habits that are important for success in the technological design industry, as identified in the Ontario Skills Passport (e.g., *working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy*);
- D2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in technological design (e.g., *work logs, skills checklist, sketches, drawings, photographs of models and prototypes*), and explain why having a current portfolio is important for career development and advancement.